HDAG:
A Hyperdimensional DAG Framework for Semantic Resonance Systems

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Abstract

This work introduces HDAG, a hyperdimensional directed acyclic graph model designed to encode semantic field architectures in the context of 5th-order cybernetics. HDAG extends conventional DAG models by embedding each node and edge into a tensorial 5D space characterized by morphodynamic resonance, phase alignment, and pre-symbolic intentionality. The result is a framework where meaning is not stored but formed, and where cognition emerges from structure rather than execution.

1 1. Introduction

Classical DAGs serve as symbolic pipelines for data processing. In contrast, HDAG redefines these graphs as semantic field infrastructures, embedding each element into a 5-dimensional resonance lattice. Here, nodes are semantic vortices, edges are phase paths, and cycles are eliminated not by design, but by phase disalignment. The HDAG system offers a fundamentally new model for field-driven machine cognition.

2 2. Theoretical Basis: 5D Tensor Graphs

Each HDAG node is represented as a resonance tensor $T_i \in R^5$, where each axis encodes a semantic dimension: temporal, morphic, relational, topological, and entropic.
Edges E_{ij} are represented as phase-gradient transitions $\Phi_{ij}(t)$, allowing the propagation of resonance and coherence. The system evolves as a tension field rather than a flowchart.

3 3. Formal Model

Let $G = (V, E)$ be an HDAG, where:

- $V = \{T_i\}$ are node tensors in R^5
- $E = \{\Phi_{ij}\}$ are directed semantic transitions

The semantic coherence condition requires that:

$$\forall (i, j) : \quad \Phi_{ij}(t) = \nabla_\mu T_i(t) \Rightarrow T_j(t + \delta t) \in S_{\text{coh}}$$

Where S_{coh} defines a phase-locking subspace within the 5D field.

4 4. Application in GRAIL Systems

In GRAIL-based architectures, HDAG operates as the substrate of structural agency. It interacts with modules like RIF, Compiler, and TINTA to orchestrate distributed intention, coherence-based task flow, and morphic field memory. HDAG's role is comparable to the 'semantic nervous system' of modular cognitive agents.

5 5. Conclusion

HDAG provides a scalable, non-symbolic graph framework for emergent machine cognition. By coupling topological directionality with semantic phase resonance, it opens a new path for post-algorithmic AI: one that thinks by forming fields, not executing instructions.

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